25ECPCC304IAE-I

Seat No. : EU125

6 AUG 2026

Student Name : Harsh

St. John College of Engineering and Management, Palghar,
Maharashtra
Autonomous Institute

Internal Assessment Examinations, August 2026

B.Tech. Electronics & Computer Science - SEMESTER III

25ECPCC304 - Sensor Technology and its Applications

Time : 1 Hour

Maximum Marks : 20

John College of Engineering & Management SJCEM PALGHAR

Instructions

- All Questions are compulsory and carry equal weightage of 5 marks each.
- Assume suitable data wherever required

Q1	Attempt any 5 out of 6 Questions	Maximum marks(5)	Course Outcome	RBT Level
	a)Identify the principle used by a capacitive level sensor to detect liquid level.	1 ✓	CO1	L3
	b)Examine the role of Analyte in Biosensor.	1 ✗	CO1	L3
	c)Examine the role of photolithography in MEMS fabrication.	1 ✓	CO2	L4
	d)Examine the role of Sacrificial layer in MEMS fabrication.	1 ✗	CO2	L4
	e)Examine the role of temperature difference in the generation of voltage through the Seebeck effect.	1 ✓	CO3	L3
	f)Examine "Increased Reliability" Feature of Sensor arrays.	1 ✗	CO3	L3
Q2	a)Apply the concept of physics to explain the working principle of any two types of Position sensors. (Use suitable diagrams)	5	CO1	L3
	OR			
	b)Utilize Suitable diagram to explain the working principle of any two types of Temperature sensors.	5 ✓	CO1	L3
Q3	a)Examine the working principle of a MOS based Gas sensor with the help of a suitable diagram.	5 ✓	CO2	L4
	OR			

	b)Examine the working principle of a MEMS accelerometer with the help of a suitable diagram.	5	CO2	L4
Q4	a)Analyse the working principle of a 4–20 mA current loop system.	5	CO3	L4
	OR			
	b)Distinguish between Traditional sensors and smart sensors.	5	CO3	L4